ACADEMIX

Enterprise Learning Management System

A production-grade, full-stack educational platform built with ASP.NET Core 8 and React 19 — featuring AI-powered recommendations, real-time communication, payment processing, and multi-role dashboards.

25+	5	22+	18	4
API ENDPOINTS	USER ROLES	DB ENTITIES	THEMES	ARCHITECTURE LAYERS

ARCHITECTURE

Clean Architecture with strict dependency flow — each layer only depends on the one directly below it.

API Layer (Controllers, Hubs, Middleware) → **Infrastructure** (EF Core, Email, Payments) → **Application** (Services, DTOs, Interfaces) → **Domain** (Entities, Enums, Core Logic)

TECHNOLOGY STACK

Backend

ASP.NET Core 8 · C# · Entity Framework Core 8 · JWT Authentication · BCrypt · Serilog · Swagger / OpenAPI · API Versioning · Rate Limiting

Frontend

React 19 · TypeScript · Vite 7 · Tailwind CSS · Zustand · Radix UI · Framer Motion · Recharts · FullCalendar

Real-time & Integrations

Microsoft SignalR · WebSockets · Lahza Payments · Mailjet Email

Database & DevOps

PostgreSQL · EF Core Migrations · Docker · ngrok · Git

MULTI-ROLE SYSTEM

Student · Teacher · Admin · Accountant · Secretary

Student Portal

- Personalized dashboard & stats
- Course catalog with smart filtering
- Video lessons with timestamped notes
- Assignments & exam submissions
- Calendar & progress tracking
- Certificate download (PDF)
- Payment history & subscriptions

Teacher Portal

- Course creation & management
- Lesson content builder
- Assignment grading & feedback
- Exam & question bank management
- At-risk student detection
- Discount & subscription plans
- Real-time messaging with students

Admin & Staff

- User management (CRUD)
- Course moderation
- System settings & audit logs
- Finance: transactions & payouts
- Revenue split management
- Enrollment bulk operations
- Reports & analytics exports

CORE FEATURES

AI-Powered Recommendations

- Hybrid algorithm: collaborative + content-based filtering
- Personalized "For You" based on enrollment history
- Trending courses by popularity metrics
- "Continue Learning" & similar course suggestions
- 6 onboarding interest categories

Real-time Communication

- SignalR WebSocket-based messaging
- Direct user-to-user & course-wide chats
- Conversation approval workflow
- Message persistence & delivery indicators
- User blocking & reporting

Advanced Analytics

Payments & Subscriptions

- At-risk student detection (inactivity, low grades)
- Engagement & risk scoring algorithms
- Predicted final grade calculation
- Auto-generated improvement recommendations
- CSV export for reports

- Lahza gateway integration with webhooks
- One-time course purchases
- Recurring subscription billing (Monthly/Yearly)
- Discount codes (percentage & fixed)
- Refund & partial refund support

SECURITY & INFRASTRUCTURE

- ✓ JWT + Refresh Tokens
- ✓ CORS Policies
- ✓ Email Verification
- ✓ Soft Deletes + Audit Trail
- ✓ Role-Based Access Control
- ✓ Rate Limiting (IP-based)
- ✓ Global Exception Handling
- ✓ Request Correlation IDs
- ✓ BCrypt Password Hashing
- ✓ Security Headers Middleware
- ✓ Structured Logging (Serilog)
- ✓ EF Core Migrations

ADDITIONAL HIGHLIGHTS

🎨 Theme Engine

18 built-in themes (14 solid + 4 gradient) plus full custom theme support via CSS variables and HSL color picker. Persisted with Zustand + localStorage.

📚 Course System

Full lifecycle: Draft → Published → Archived. Supports video lessons, materials, sections, multiple modalities (Online/In-Person/Hybrid), and auto-certificates on completion.

📝 Assessment Engine

Assignments with multi-attempt submissions & file uploads. Exams with MCQ, True/False questions, timed attempts, and automated scoring with attempt history.